

IEEE 754 FLOATING POINT

ADDITION AND SUBTRACTION (Half Precision)

Project Learning Outcomes

PLO1 What is IEEE?

PLO2 What does IEEE 754 means?

PLO3 Format of Floating point representation.

PLO4 How does manual human floating point addition and subtraction carried out.

PLO5 Through Logisim file we will know how actually computer stores floating point number and how operations in different subcircuits performs.

PLO1: What is IEEE?

→ IEEE stands for Institute of Electrical and Electronics Engineers.

→ It is a big ~~organization~~ international organisation of engineers and scientists who work in technology fields like:

- Computer
- Electronics
- Communication.

What does IEEE do?

- Create standards.
- Publish research papers
- Organizes conferences & events
- Supports students & professionals

PL02: What does IEEE 754 means?

→ So IEEE 754 is a standard (rule) that tell the computers how to store and calculate floating point numbers.

→ Why do we need it?

→ Because computer is dumb it understands only 0's and 1's not like "101.1011".

→ So it is globally fixed method to represent IEEE 754 Floating point numbers.

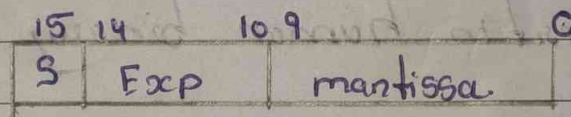
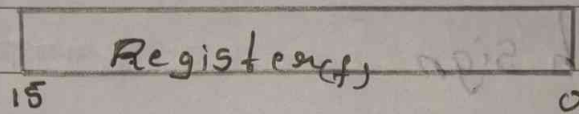
PLO3: Format of floating point representation

→ It IEEE 754 format is divided into 3 fields.

1. Sign

2. Exponent

3. mantissa (Fractional part)



(Register R_f)

$R_f(15) \rightarrow$ Sign bit : $0 = +ve$, $1 = -ve$

$R_f(10-14) \rightarrow$ Exponent: power of 2 for binary.

$R_f(0-9) \rightarrow$ mantissa: Fractional Part.

→ The final exponent and mantissa is achieved ~~there~~ after normalisation.

→ Normalisation is a process where only one significant bit is sufficient ~~for~~ must be before decimal point.

→ It can be either 0 or 1.

How to convert floating point into IEEE 754 format.

Example:

5.75

Step: 1 check sign

$$\text{Sign}(5.75) = +ve$$

Step: 2 Convert in human binary representation

$$5 = 101$$

$$0.75 \times 2 = 1.5 \rightarrow 01$$

$$0.5 \times 2 = 1.0 \rightarrow 10$$

$$0.0 \times 2 = 0 \rightarrow 0$$

$$5.75 = 101.110$$

Step: 3 Normalise.

$$101.110 \times 2^0$$

$$1.0111 \times 2^2$$

Step: 4 Find each field

$$\text{Sign} = +ve$$

$$\text{exponent} = \text{bias} + \text{actual exponent of 2}$$

$$\text{exponent} = 15 + 2 = 17$$

mantissa = the numbers in binary after decimal point

Why do we need Bias? What its significance?

→ Sometimes we have -ve exponents but we know our computer stores directly 1's and 0's

→ It cannot directly represent negative numbers (without any extra complexity like 2's complement)

→ Solution is just add bias just to make the negative exponents to positive.

Advantages

→ Easy for computer to identify when bigger exponents.

→ No need for handling in exponent.
(Hardware simplicity).

$$\text{Bias} = 2^{(K-1)} - 1$$

where K = no. of exponents bits.

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So the exponent value is represent in between

→ The Bias is 15 because $2^5, 32$ values 15 for negative and 15 for positive

Here in 5 bits we have range

$00000 \rightarrow 0$
 $11111 \rightarrow 31$ } 32 numbers

$00000 \rightarrow 0$ and $11111 \rightarrow 31$ are used for special cases.

So the usable range is

$00001 \rightarrow 11110$

1 to 30 = E

Now the actual exponent = $E - 15$ } for having represented

So it becomes

-14 to 15

Smallest positive Number

(A) Smallest Normal Number

When $exp = 00001$

Fraction is $= 0000000000$

$= 1 \times 2^{-14}$ } in binary

value $\frac{1}{2^{14}} = 0.000061035$

$= 6.1035 \times 10^{-5}$

(3) Smallest Denormal Number

Exponent = 00000

Fraction = 0000000001

$$= 0.00000000000001 \times 2^{-14}$$

$$\text{value} \approx 5.96 \times 10^{-8}$$

Smallest positive is 5.96×10^{-8}

2. Largest Number

Exponent = 11110

Fraction

Fraction = all 1's

$$1.11111111$$

$$\text{value} = 1.11111111 \times 2^{15}$$

$$= 1.999023 \times 2^{15}$$

$$= 65504$$

Final range we can use in this half precision

$$\pm (5.96 \times 10^{-8}) \text{ to } \pm 65504$$

How to convert IEEE format to back in to human decimals.

1) For normal numbers

$$\text{Value} = (-1)^{\text{Sign}} \times (1 + \text{fraction}) \times 2^{(\text{Exponent} - \text{bias})}$$

2) For denormal numbers

$$\begin{aligned} \text{Value} &= (-1)^{\text{Sign}} \times (0 + \text{fraction}) \times 2^{(1 - \text{bias})} \\ &= (-1)^{\text{Sign}} \times (0 + \text{fraction}) \times 2^{-14} \end{aligned}$$

Examples.

1) 0100010100000000

$$\begin{aligned} \text{Sign} &= 0 \text{ +ve.} \\ \text{exponent} &= 17 \\ \text{actual} &= 17 - 15 = 2 \\ \text{mantissa} &= 0100000000 \end{aligned}$$

$$\text{Value} = (-1)^0 \times (1.0100000000) \times 2^2$$

$$= (1 + 0 + 0.25) \times 4$$

$$\begin{aligned} &= 1.25 \times 4 \\ &= 5 \end{aligned}$$

2) 0000000100000000

Sign = 0 $\rightarrow$ +ve
 exponent = 00000
 actual = -14

$$\text{Value} = (-1)^0 \times (0.0100000000) \times 2^{-14}$$

$$= \frac{0.25}{2^{14}} = 0.000015258$$

$$= 1.52 \times 10^{-5}$$

Special cases. that are designed by IEEE Organisation

1) Zero (Underflow)

Sign = $\pm$ ve
 exponent = 00000
 mantissa = 0000000000
 +0 000000000000000000
 -0 100000000000000000

2) denormal numbers. (Underflow)

$\rightarrow$ These are the numbers that are close to zero

Sign = $\pm$ ve
 exponent = 00000
 mantissa = anything.

0 0000001000000000

3) Infinity (Overflow)

Sign = $\pm Ve$
exponent = 1111
mantissa = 000000000000

$+\infty$ 0 1111 000000000000
 $-\infty$ 1 1111 000000000000

B) NaN (not a number) (Overflow)

Sign = $\pm Ve$
exponent = 1111
mantissa = ~~000~~ anything

0 1111 100100000000

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PLO4

How, addition and Subtraction are carried out?

ADD

Step 1: Convert both numbers into IEEE format

Step 2: Make exponents equal

→ Then shift the number that has lower exponent value

Step 3: Add mantissas
like normal binary addition

Step 4: Normalise result
Convert into

$$1.xxxxxx \times 2^n \text{ or } 0.xxxxxx \times 2^n$$

Step 5: Adjust exponent

Step 6: convert back into IEEE format.

Example

$$5.5 + 2.5$$

$$\begin{aligned} 5.5 &= 101.1000000000 \times 2^0 & 0.5 \times 2 &= 1.0 \quad 0 \\ &= 1.0110000000 \times 2^1 & 0.0 \times 2 &= 0 \end{aligned}$$

$$\begin{aligned} 2.5 &= 10.1 \times 2^0 \\ &= 1.0100000000 \times 2^1 \end{aligned}$$

$$0.1010000000 \times 2^2$$

$$5.5 + 2.5 =$$

$$1.0110000000 \times 2^2$$

$$0.1010000000 \times 2^2$$

$$\hline 10.0000000000 \times 2^2$$

$$= 1.0000000000 \times 2^3$$

$$= 8 \times 8$$

$$= 8$$

8

Subtraction

(A-B)

All steps are same we just have to flip the sign of B

$$\Rightarrow A + (-B)$$

$$5.5 - 2.5$$

$$5.5 \rightarrow 1.0110000000 \times 2^2$$

$$2.5 \rightarrow 0.1010000000 \times 2^2$$

$$-2.5 = -0.1010000000 \times 2^2$$

5.5

$$\begin{array}{r} 0.10 \\ 1.0110000000 \times 2^2 \\ + 0.1010000000 \times 2^2 \\ \hline 1.1100000000 \times 2^2 \end{array}$$

$$= 1.1100000000 \times 2^2 \Rightarrow 0.10000000000000000000$$

$$= (1 + 0.5) \times 2 = 3$$

$$= 1.5 \times 2$$

$$= 3$$

What if ~~A-B~~ and $-A+B$

then represent into $B-A$
then change sign of A then add.

For $-A-B$

→ First do $(-A) + (-B)$

→ Flip sign of both A then add
The sign is negative B

For ex

$$\Rightarrow -6-2$$

$$A=6 \quad B=2$$

$$A \rightarrow 110.0000000000 \times 2^0$$

$$= 1.1000000000 \times 2^1$$

$$B \rightarrow 10.0000000000 \times 2^0$$

$$= 1.0000000000 \times 2^1$$

$$-A = 1100011000000000$$

$$-B = 1100000000000000$$

$$\begin{array}{r} 11 \\ -1.1000000000 \times 2^2 \\ -0.1000000000 \times 2^2 \\ \hline -10.0000000000 \times 2^2 \\ -1.0000000000 \times 2^3 \end{array}$$

$$\begin{array}{r} -10.0000000000 \times 2^2 \\ -1.0000000000 \times 2^3 \end{array}$$

$$= -8$$